AGPS 1024 · PRINCIPLES OF PLANT SCIENCE · LAB

DESIGN PROJECT: THE SCOTT RESIDENCE

Designer (your name) | Lab section | Due date

For this project you are a landscape designer, and the Scott family has hired you to design the yard at their new home. This handout is your notes from the client interview. You will study the site, design the landscape, turn in your plans, and then present your design to your client. **On presentation day, the class plays the Scott family**, so design it for them. They will be the ones asking the questions.

THE JOB

Client	The Scott family (two parents and three teenagers)
Site	New two-story home on a completely cleared lot. See the base plan.
Your role	Landscape designer
You deliver	Site analysis, functional diagram and a general design plan
Final step	Present your design to the client (the class) and answer their questions

WATCH A DESIGNER AT WORK

Before you start, we will watch landscape designer Lisa Nunamaker talk about how she designs for her clients (youtube.com/watch?v=SyR5wBboHrA). Watch it as if you are a new designer learning the job, because next you will be doing the same thing for the Scotts. Answer these as you watch.

1. How does she find out what her clients want? What kinds of questions does she ask them?

2. What does she look at on the site before she starts designing?

3. How does she show her design to the client? What does she use (drawings, photos, plans)?

4. Give one example of a plant choice or placement she explains. Which plant science idea from lecture is behind it?

5. Name one thing she does that you will use in your design or presentation for the Scotts.

MEET YOUR CLIENT

Here is what the Scotts told you in your first meeting.

FAMILY MEMBER	WHAT THEY TOLD YOU
Husband, 40	Enjoys gardening; likes fruit trees, especially apple; plays with the kids in the yard; entertains with outdoor cookouts
Wife, 40	Enjoys vegetable and flower gardening; loves to cook and host small groups for dinner
Twin boys, 15, and **daughter**, 13	Active in sports and want an open lawn; love nature, especially bird-watching

WHAT THE CLIENT WANTS

Your design has to deliver every one of these. Check each off as you put it on your plan.

- [ ] Sidewalk to the front door
- [ ] Recreation area
- [ ] Outdoor sitting and entertaining area with partial shade
- [ ] Privacy on the patio, and from the busy street on the west side
- [ ] Flower gardens, front and back
- [ ] Small vegetable garden
- [ ] Wood stockpile area

WHAT YOU KNOW ABOUT THE SITE

- House is a two-story traditional with light grey siding and white trim.
- Busy street on the west side of the house.
- The family places a high value on getting along with their neighbors.
- Cold winds blow from the north; hot winds blow from the south.
- Overhead electric runs along the back (east) property line at 15 ft, with a service drop to the house at 12 ft. An underground line runs from the southwest corner of the house to the street, 2 ft below grade.

WHAT YOU TURN IN

Turn these in together, with your name on each sheet, at the start of **next lab**. You will present your design to the client in that same lab, so come ready to present.

1. **Site analysis:** your answers to Step 1, with what you found marked on the base plan.
2. **Functional diagram:** on tracing paper over the base plan (Step 2).
3. **Design plan:** drawn on the design plan sheet, every plant labeled by type, title block filled in (Step 3).

You will get two 11×17 sheets: the **base plan**, a reference copy with dimensions, utilities and the symbol key (1 in = 10 ft), and a blank **design plan** sheet for your final design. You will also need tracing paper, an H or HB pencil, a 2H or 4H pencil for guidelines, an engineer's scale, a circle template, and the reference sheets in your packet (lettering guide, plan graphic symbols, and the Duncan residence examples).

HOW DESIGNERS WORK

Landscape design is a creative problem-solving process. The goal is to organize outdoor space so the natural features of a site and the needs of the people using it are in balance: plants and built elements (walks, patios, fences, structures) arranged so a home or business works well and looks good (adapted from G.M. Pierceall, *Residential Landscapes*). Landscape designers, landscape architects, interior designers and architects all follow about the same process:

#	STEP	WHAT HAPPENS
1	**Research and preparation**	Interview the client, prepare a base map, do a site inventory and analysis. **Done for you: the interview and base plan. You do: the site analysis.**
2	**Design**	Functional diagram, then the general design. **You do this.** Your presentation is the client meeting where you show it.
3	Construction documents	Layout plan, grading plan, planting plan, construction details.
4	Implementation	Construction and installation.
5	Maintenance	Pruning, mowing, irrigation, replacement.
6	Evaluation	Did the design do what the client needed?

A good design solves the client's problems without creating new ones. A shade tree planted under a power line, or a hedge that blocks the view of traffic from the end of the driveway, both "solve" something and cause a bigger problem. Before you draw anything, account for:

1. How the client wants to use the space
2. Budget
3. Plant selection (see Design reference below)
4. Soil, rainfall, climate and other site conditions
5. Existing wildlife
6. Ordinances, permits, HOA rules and utility setbacks

STEP 1 — STUDY THE SITE

Before you design anything, you need to know the site. Answer these here, mark what you find on the base plan in light pencil, and turn this page in with your design.

1. Label all four sides of the lot N, S, E and W. Which side of the house gets hot afternoon sun? Which side takes the cold winter wind?

2. Find the overhead and underground utility lines. Shade the parts of the lot where you should *not* plant a large tree, and explain why.

3. The finished floor is EL. 25.0, the ground at the front door is EL. 23.5 and the ground at the back is EL. 24.0. Does surface water drain toward or away from the house? What does that tell you about where to put the vegetable garden?

4. Which doors open from the rooms the family uses to cook and entertain? Circle them. Where does that suggest the patio should go?

5. Find the A/C unit, the gas meter and the garage side door. Why do these belong in or near the service area?

6. Predict: if you plant a solid row of evergreens along the whole west property line, what happens to (a) afternoon sun on the front of the house, and (b) the view of oncoming traffic when the family backs out of the driveway?

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7. Why should the wood stockpile not sit right against the house?

STEP 2 — FUNCTIONAL DIAGRAM

Tape a sheet of tracing paper over the base plan. Draw loose "bubbles" for each use area and label them. Do not draw plants yet. The Duncan residence diagrams in your packet show what this should look like.

- **Public:** front yard and entry walk
- **Private:** patio and entertaining area, sitting area
- **Service:** wood stockpile, A/C, gas meter, trash
- **Recreation:** open lawn for sports
- **Garden:** vegetable garden, flower beds, fruit trees

Put each area next to the door that serves it, and draw arrows to show how people move between doors and areas. Show me your diagram before you start Step 3.

STEP 3 — GENERAL DESIGN PLAN

On the design plan sheet, draw the design you will present to the Scotts. Work it out on tracing paper first if you want, then draw the final version on the sheet.

1. Hardscape first: walk to the front door, patio, paths, wood stockpile.
2. Then trees, then shrubs, then beds, ground cover and lawn.
3. Draw every plant at its **mature spread**, in proportion to the house. The house is about 41 ft across, so a tree with a 30 ft spread should be about three-quarters as wide as the house.

4. Label every plant by type (deciduous tree, evergreen shrub, herbaceous perennial, and so on). You do not need plant names.
5. Use the plan symbols from the key on the base plan, and letter labels in capitals between light guidelines.
6. Fill in the title block with your name, lab section and date.

DESIGN REFERENCE

Use this as a checklist while you design. It all comes back to what we covered in lecture: mature plant size, deciduous vs. evergreen leaf habit, and how plants change light, temperature and air movement.

TREES

- **Right tree, right place.** Plan for height and spread at maturity, not the size at planting.
- **Scale.** Large trees dwarf a small house; they add distinction to a large one.
- **Spacing.** 15–20 ft from the side of the house, 12–15 ft from a corner, and large trees no closer than 25 ft to a septic tank, sewer or other utilities.
- **Uses.** Visual dividers; a leafy roof over a sitting area; shade that cools the spot below; screens that hide an unsightly area. Evergreens screen year-round and cut down on noise. Two trees can frame a view.
- **Qualities to look for.** Strong wood; disease and insect resistance. Lawn trees should have fine texture so light gets through to the turf.
- **Winter sun, summer shade.** Put deciduous trees on the southwest side.

WIND PROTECTION

- Windbreaks block and redirect the flow of wind.
- In our region, winter winds come mostly from the north or northwest and summer winds from the south or southwest. For the Scott site, use the wind directions the client gave you.
- A windbreak gives its best protection for a distance of about 5–7 times its height downwind.

SHRUBS

- Practical on small lots where there is no room for big trees.

- Plant close enough together to form a solid wall, but far enough from the house to leave a dead-air space. Still air insulates better than moving air.
- Rows of shrubs on the northwest side protect the house from cold winter wind as well as direct summer heat.

ESPALIERS AND VINES

An espalier is a plant trained to grow flat against a wall or frame.

- Shades walls and windows in summer and buffers the wall from wind in winter.
- Deciduous types let sun reach the wall in winter.
- Vines can be used as ground cover or trained on an arbor (American wisteria, Carolina jessamine).
- Overhead structures such as arbors and slatted wood shade structures do the same job where there is no room for a tree.

PLANT TYPES FOR YOUR PLAN

Label every plant with one of these types, and be ready to tell your client why that type fits that spot.

PLANT TYPE	WHAT IT DOES WELL
Deciduous (tree or shrub)	Summer shade, winter sun; seasonal flowers, fruit and fall color
Evergreen (conifer)	Year-round screening, windbreaks, noise reduction
Broadleaf evergreen	Year-round foliage at a smaller scale: foundation plantings, low screens, hedges
Herbaceous perennial	Flower beds that come back every year; die back to the ground in winter
Annual	Seasonal color, replanted each year; small, high-visibility spots near entries and the patio

Symbols for each type are in the key on the base plan.

LETTERING REMINDERS

- Keep the pencil point sharp and roll it as you go so the line weight stays even.
- Press a little harder at the start and end of each stroke.
- Always draw light guidelines first with a 2H or 4H pencil; vertical strokes touch both lines.

- Space between lines of lettering should be about half the letter height.

PRESENTING TO YOUR CLIENT

On presentation day the class is the Scott family. You get 3–5 minutes to present your design, then the client gets to ask questions.

WHEN YOU PRESENT

- Put your plan where the client can see it and point to what you are talking about.
- Go through the client's list. Show where each item is and why you put it there.
- Explain how your design handles sun, wind, the utility lines and the busy street.
- Talk about what the family gets ("shade on the patio by late afternoon"), not about drawing symbols.

WHEN YOU ARE THE CLIENT

When you are in the audience, you are the Scotts. Ask the questions a homeowner would ask, for example:

- Where do the kids play ball, and will the ball end up in the garden?
- Will we have shade on the patio when we eat dinner in July?
- Can the people driving by see into our patio?
- What will the front yard look like in January?
- Where do the birds come in?

HOW YOUR DESIGN WILL BE GRADED

The question is simple: did you deliver what your client asked for?

CRITERIA	POINTS
Client wants and needs. Every item on the client's list is on the plan, sized and placed sensibly.	15
Site conditions. Design responds to sun, wind, drainage, utilities and the street.	10

CRITERIA	POINTS
Plant selection and placement. Right plant type in the right spot, drawn at mature size, proper distance from the house.	10
Site analysis and functional diagram. Step 1 answered; use areas make sense and circulation from the doors works.	5
Graphics. Correct symbols, every plant labeled by type, neat lettering, title block filled in.	5
Client presentation. Explains the design in terms the client cares about and answers the client's questions.	5
Total	**50**